Qatar-1b

R-0463

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_170629` · created 2026-07-19T21:33:38+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457933.87365 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.01
Transit depth (Rp/Rs)^2	3.68 +/- 0.39 [%]
Orbital Inclination (inc)	83.39 +/- 0.59
Airmass coefficient 1 (a1)	1.087 +/- 0.017
Airmass coefficient 2 (a2)	-0.063 +/- 0.073
Scatter in the residuals of the lightcurve fit is	1.52 %
Best Comparison Star	#2 - [271, 83]
Optimal Aperture	2.87
Optimal Annulus	5.91
Transit Duration (day)	0.0694 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.01 vs 0.1463 (deviation 4.43σ)
Duration fit vs published	0.0694 d vs 0.0692 d (deviation 0.06σ)
Mid-time O–C	1.8 min
Depth SNR	18.6
Residual scatter	1.52 %

Light curve

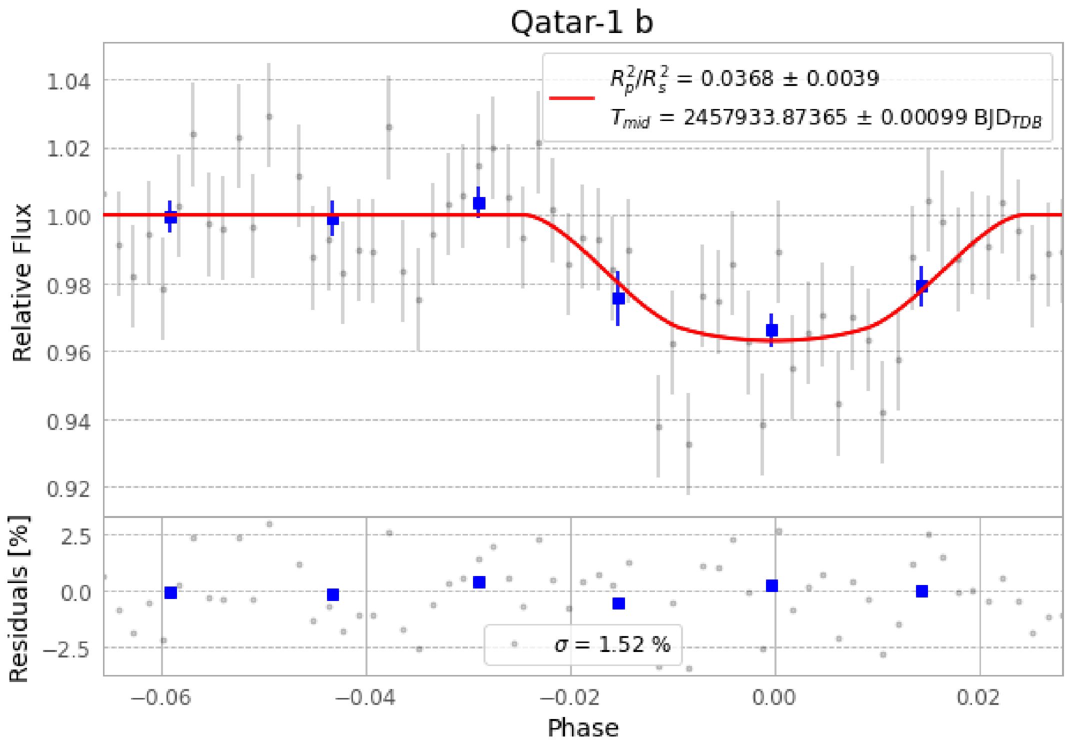

Figure 1 — FinalLightCurve_Qatar-1 b_2017-06-28.png (SHA-256 90cc9e05bbcfed2e…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 237], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 70916cf7bfe08fba…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

65 FITS frames (43 MB), `Qatar-1170629064212.FITS` → `Qatar-1170629095412.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-170629.log (064eb53c921f…), FinalLightCurve_Qatar-1 b_2017-06-28.png (90cc9e05bbcf…), FinalLightCurve_Qatar-1 b_2017-06-28.csv (27569be19918…)

Compute resources

Wall time 310.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 21:33Z.